

Modes of DES (ECB, CBC, CFB, OFB)

1. Aim

To demonstrate the four basic modes of operation of DES - Electronic Code Book (ECB), Cipher Block Chaining (CBC), Cipher Feedback (CFB) and Output Feedback (OFB) - and to study how an initialization vector influences the encrypted output in each mode.

2. Algorithm

1. Start the program.
2. Declare variables for plaintext, key, initialization vector (IV) and the four result variables ecb, cbc, cfb and ofb.
3. Read the plaintext, key and initialization vector entered by the user.
4. Compute the ECB result by performing XOR of plaintext directly with the key.
5. Compute the CBC result by performing XOR of plaintext with the IV first, then XOR the result with the key.
6. Compute the CFB result by performing XOR of plaintext with the XOR of IV and key.
7. Compute the OFB result by performing XOR of plaintext with the XOR of key and IV.
8. Display the ECB, CBC, CFB and OFB values and stop the program.

3. Program

```
#include <stdio.h>

int main() {
    int plaintext, key, iv;
    int ecb, cbc, cfb, ofb;

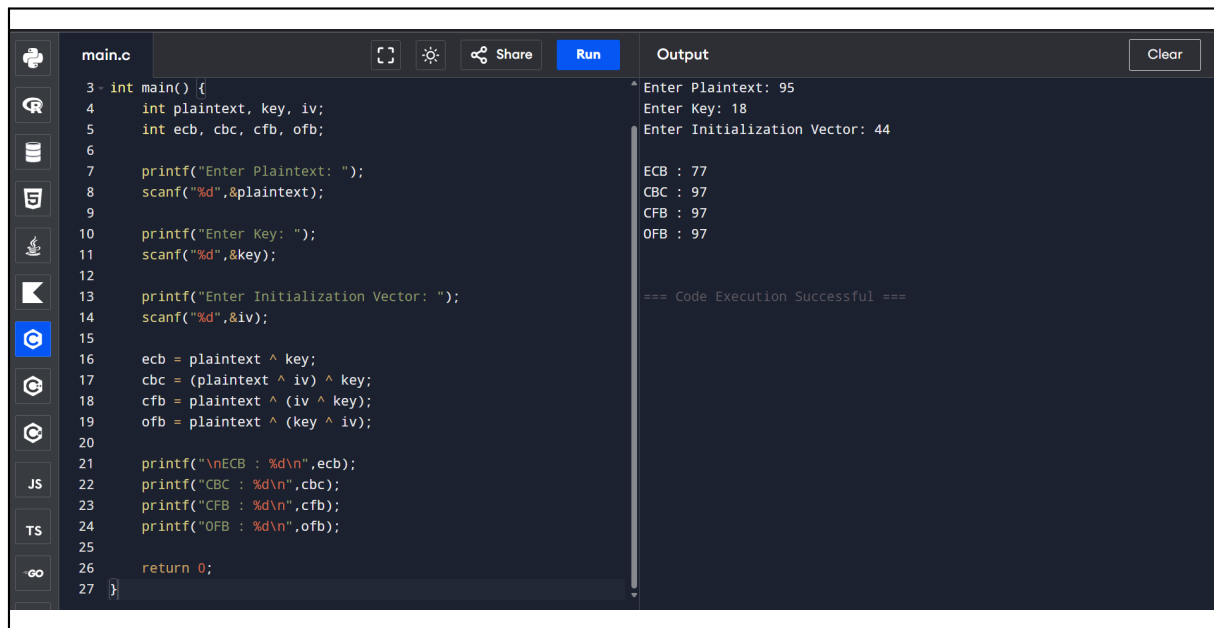
    printf("Enter Plaintext: ");
    scanf("%d", &plaintext);
    printf("Enter Key: ");
    scanf("%d", &key);
    printf("Enter Initialization Vector: ");
    scanf("%d", &iv);

    ecb = plaintext ^ key;
    cbc = (plaintext ^ iv) ^ key;
    cfb = plaintext ^ (iv ^ key);
    ofb = plaintext ^ (key ^ iv);

    printf("\nECB : %d\n", ecb);
    printf("CBC : %d\n", cbc);
    printf("CFB : %d\n", cfb);
    printf("OFB : %d\n", ofb);

    return 0;
}
```

4. Output Screenshot



The screenshot shows a code editor with a file named `main.c` and an output window. The code implements a DES encryption program with ECB, CBC, CFB, and OFB modes. The output window shows the results of a successful execution with the following inputs and outputs:

```
main.c
3 int main() {
4     int plaintext, key, iv;
5     int ecb, cbc, cfb, ofb;
6
7     printf("Enter Plaintext: ");
8     scanf("%d",&plaintext);
9
10    printf("Enter Key: ");
11    scanf("%d",&key);
12
13    printf("Enter Initialization Vector: ");
14    scanf("%d",&iv);
15
16    ecb = plaintext ^ key;
17    cbc = (plaintext ^ iv) ^ key;
18    cfb = plaintext ^ (iv ^ key);
19    ofb = plaintext ^ (key ^ iv);
20
21    printf("\nECB : %d\n",ecb);
22    printf("CBC : %d\n",cbc);
23    printf("CFB : %d\n",cfb);
24    printf("OFB : %d\n",ofb);
25
26    return 0;
27 }
```

Output:

```
Enter Plaintext: 95
Enter Key: 18
Enter Initialization Vector: 44

ECB : 77
CBC : 97
CFB : 97
OFB : 97

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter Plaintext: 77
Enter Key: 9
Enter Initialization Vector: 5

ECB : 68
CBC : 65
CFB : 65
OFB : 65
```

6. Result

The program demonstrating the ECB, CBC, CFB and OFB modes of DES was implemented and executed successfully, and the effect of the initialization vector on each mode's output was observed.